

BLUE WATER

Even in shallow seawater, everything looks blue or blue-green. This is because all the other colors in sunlight have been absorbed by the water. Red light is absorbed first, followed by yellow, green, and violet, until only blue is left. Any light reflected up through the water has also lost these colors, which is why the sea usually looks blue.

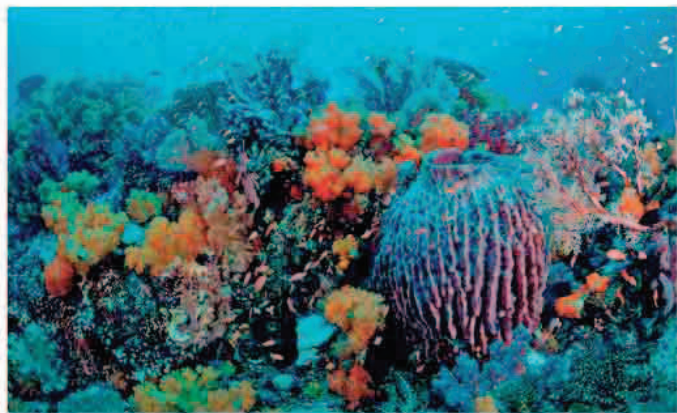


▲ NATURAL LIGHT

This photo of a coral reef shows it lit by natural blue-green underwater light.

▼ FLASH LIGHT

The pure white light of a camera flash reveals the true colors of the coral reef.



Light, heat, and sound

Water absorbs light and heat. This means that neither can penetrate far into the ocean depths—unlike sound, which travels well through water. Ocean water is also slow to warm up and cool down. This has a big effect on the climates of nearby shores, and it also allows ocean currents to carry heat to other parts of the globe.



OCEAN TEMPERATURES

The Sun shines straight down on the tropics, warming the surface waters of tropical oceans to temperatures of up to 86°F (30°C). But in the polar regions, the Sun is lower in the sky and less powerful, even in summer. In winter there is so little warm sunlight that the seas freeze over. But warm currents flowing toward the poles stop the sea from getting even colder, and cold currents flowing away from polar seas help to cool the tropics.

